A Retrospective on the DCHL Substrate: Which Mechanisms Survived a Retired Architecture, and Why

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> *"A dead paper is not a retracted one. Four of the mechanisms running in production today belong to papers filed as retired."*

Abstract

Between 2025 and 2026 this lineage produced a family of continuous-state cognitive architectures — CMA, P-CMA, CCH, HCMA/DCHL, CMC, ANQ, ZeroGrad, BED, CME — organised around

a learned potential Φ on a cognitive manifold. **That substrate is retired.** It collapsed into the failure modes its successor formalises as the classifier, memorisation and retrieval traps.

This paper is the salvage record: what survived, what did not, and why the two questions have different answers than the folder structure suggests. Four mechanisms whose papers are filed as retired are load-bearing today — **ZeroGrad**'s local Hebbian rule, **ANQ**'s continuous-integrator pattern, **CMA**'s mutation primitive, and **LTO**'s runtime acquisition. One constraint that gave the lineage its name — **CCH**'s curl-free field — was never implemented, although the *geodesic* half of the same paper found a home in a place its author did not anticipate.

Negative results are reported as results. HCMA's diagnosis of hidden-layer collapse under a global scalar teaching signal, and its observation that a static potential cannot traverse a sequence, were correct, were derived rather than guessed, and motivated the oscillator substrate that replaced them.

Every status claim was verified by reading the live source, not the paper that made it.

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1. Why Salvage Deserves a Paper

A research lineage accumulates claims faster than it retires them. Papers are written when work begins to succeed and are rarely revisited when it stops. The result is a literature describing a system nobody runs.

Two failure modes follow. **Over-claiming**: a synthesis section written once and never re-checked keeps listing components removed long ago. **Under-claiming**: a mechanism is buried with the architecture it was introduced in, even though the mechanism was sound and only the architecture failed.

The second is the more expensive error. The unit of retirement should be **the proposal**, not **the idea**.

1.1 Method, and a warning it produced

Each status claim was produced by searching live source for the mechanism, then reading the code to confirm the match was not incidental. Where paper and source disagreed, the source won.

The method immediately surfaced a hazard worth stating on its own:

> **Two distinct papers in this lineage are both called CMA.**
>
> - `CMA.md` — *Conceptual Mutation Architecture.* **Live**, implemented in `src/bisociation.hpp`, which names the paper in its first line.
> - `failed_but_parts_works/cma.md` — *Cognitive Manifold Architecture.* Retired with the continuous-state family.
>
> An audit performed by filename rather than by content will conclude that CMA is retired, which is false for one of the two. This is recorded because it is exactly the class of error the audit exists to catch, and it nearly escaped.

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2. The Substrate That Was Retired

The retired family shared one commitment: **cognition as motion in a continuous state space under a learned potential.**

$$F(c) = -\nabla\Phi(c), \qquad \dot{c} = \frac{\partial H}{\partial p}, \qquad \dot{p} = -\frac{\partial H}{\partial c}$$

A concept was a point c on a manifold; reasoning was a trajectory; logical consistency was to be enforced by making the field conservative. Training was contrastive: push Φ down on coherent states, up on incoherent ones.

The commitment failed. Sections 3 and 4 separate the parts of the failure that were *diagnosed correctly* from the mechanisms that were *worth keeping*.

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3. Negative Results Worth Keeping

3.1 Hidden-layer collapse is mathematical, not a tuning bug

HCMA identified that the contrastive Hebbian update for $\Phi(c)$ used a **global scalar** energy signal applied uniformly to every hidden unit. Every unit received the same gradient direction, so none could specialise.

The important part is the modality of the diagnosis:

> These issues are not mere engineering bugs — they are **mathematical consequences** of using a single scalar potential both to represent "goodness" and to drive generation.

This is the reasoning pattern the successor lineage later formalised as the difference between a tuning problem and a category error, and it was applied correctly here first. HCMA's own response — per-unit contrastive errors $h_j^{\text{pos}} - h_j^{\text{neg}}$, i.e. **decoupling** the teaching signal — was the right shape, which is why the substrate is named DCHL.

Status: the diagnosis survives; the fix did not save the substrate. Decoupling removed the collapse and left §3.3 untouched.

3.2 A static potential cannot traverse a sequence

A single particle in a static potential falls into the nearest local minimum and stays. Sequence generation requires either a time-dependent landscape or an energy-injection mechanism.

This is a correct and general statement about gradient flow, and it is the direct ancestor of two current mechanisms: driven oscillation — a cavity is *driven*, not released into a basin — and the chaotic metastability work that attacks ground-state locking.

Status: fully survives, restated in oscillator terms.

3.3 Contrastive training on coherence is classifier training

A system trained to separate coherent from incoherent states learns an energy function that discriminates two classes. At inference the state descends into the winning class's centroid. Calling the descent "physics" does not change what was learned.

Measured on the retired substrate at 539 M parameters (D = 512, H = 1 048 576, 4-bit ANQ, RTX 4060 8 GB): a 4-epoch pass over 8 000 C4/Pile samples in 4 min 43 s, with correctly-signed denoising $\Delta\Phi$ but hidden activations saturating at $|h| \approx 0.008$ — deep in tanh's linear regime, traceable to an aug-norm scaling of $\sqrt{\|c\|^2 + 1}$ compressing pre-activations by $\Theta(1/\sqrt{D})$.

A fix was identified (divide by $\sqrt{(\|c\|^2 + 1)/D}$) and **never propagated**, because the substrate was retired rather than repaired. This is recorded so the number is not mistaken for a live benchmark.

Status: this result is the evidence base for retiring the family. It is the most useful thing the substrate produced.

---

4. Mechanisms That Outlived Their Papers

4.1 ZeroGrad — the paper is dead, the rule is foundational

ZeroGrad proposed eliminating backpropagation entirely: each component trained by a **local, forward-only** objective with Hebbian updates, extending Forward-Forward-style local learning to the P-CMA components.

The architecture it trained is gone. **The rule is live**, carried verbatim:

$$\Delta w_{ab} \;=\; w_{ab}\,\gamma_{\text{decay}} \;+\; \eta\, v_a v_b$$

Every Hebbian update in the current engine is this rule, including the phase-consistent extension gating consolidation on $\cos(\theta_a - \theta_b - \Psi)$. It survived a change of substrate because it never depended on one: it is local, needs only two adjacent activations, and assumes nothing about what the state space means.

Status: LIVE, foundational.

4.2 ANQ — one idea survived the quantisation scheme

ANQ was 4-bit weight quantisation preserving DCHL physics, retired with the GPU trainer it served. There are no weights left to quantise.

One idea survived: **replace threshold-and-commit with a continuous integrator.** Rather than accumulating evidence until a threshold fires and then discretely committing, maintain a saturating accumulator that never makes a hard decision. Every saturating quantity in the current tick — salience, tension, endurance, identity, thermal habituation — follows:

$$x \;\leftarrow\; \tanh(x\,\lambda + \eta\,u)$$

The payoff is structural rather than numerical: a system with no thresholds has no branches, and the current tick loop contains exactly one branch — a null guard — in its entire update path.

Status: scheme DEAD, pattern LIVE and pervasive.

4.3 CMA (Conceptual Mutation) — live, and nearly lost to a filename

The Mutation primitive forces a hypothesised edge between two semantically **distant** nodes, $d(a,b) > \tau_{\text{novelty}}$, then enters a resolution phase searching for a bridge subgraph that would explain the connection. If a bridge is found within budget and the hypothesis energy is below U_{crit}, the edge is adopted; otherwise the hypothesis decays and is discarded.

The discard path is the load-bearing detail. A mutation mechanism that kept its failures would inject permanent noise into the graph; because unvalidated hypotheses decay, the substrate can afford to guess aggressively.

This is distinct from repurposing search, which looks for candidates that already *share* a mechanism target. Mutation deliberately forces pairs that share **nothing** and asks whether a bridge can be invented.

Status: LIVE (`src/bisociation.hpp`, which cites `CMA.md` in its first line). See §1.1 for why this was nearly filed as retired.

4.4 LTO — dead binary, live principle

The LTO **trainer** is retired with the DCHL substrate it updated, and its source now sits in `legacy/`.

The LTO **principle** — acquiring knowledge at runtime and persisting it so the next boot inherits it — is live in the axiom loader, which cites `lto.md` §2.2 by name for its streaming-update semantics. The mechanism changed completely: instead of updating a learned potential online, the system appends a typed triple to a file. The commitment survived; its implementation did not.

Status: architecture DEAD, principle LIVE.

4.5 RIFA — never actually retired

RIFA's Integrity Core and Silence Integration were filed with the partly-working set, but both are load-bearing:

$$\text{tension} \leftarrow \text{tension} \cdot \gamma + \beta \sum \tfrac{1}{2} m v^2, \qquad \text{coherence} = \sigma(\text{tension}/2)$$

Contradiction metabolism and identity persistence run on these. The filing was clerical, and the paper is duplicated byte-identically in two locations — itself a symptom of the bookkeeping problem this paper addresses.

Status: LIVE. Should never have been in the salvage folder.

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5. The Instructive Case: CCH, Half Dead

CCH gave the lineage its central theorem. For a conservative field $\mathbf{F} = -\nabla\Phi$:

$$\nabla \times \mathbf{F} = 0 \quad \Longrightarrow \quad \int_A^C \mathbf{F} \cdot d\mathbf{s} \text{ is path-independent}$$

The intended reading: **logical transitivity as a geometric guarantee**. The argument is mathematically correct. It is also **not implemented** — a search of the live substrate finds no `curl`, no enforcement term, and no test that the field is conservative. The constraint was stated, never imposed.

5.1 What happened to the transitivity goal

The goal was achieved by **leaving the continuous domain**. Transitivity is now a typed rewrite rule that fires exactly:

```
(X is_a C) ∧ (C is_a D)  ⊢  (X is_a D)      [premises cited]
```

The derived edge records the two premises that produced it, so transitivity is not merely guaranteed — it is *auditable*, which the field formulation never offered. One can print the proof.

> A guarantee you can prove about a mechanism you did not build is worth less than a weaker property you can print from the mechanism you did.

5.2 What happened to the geodesic framing — the part I got wrong

An earlier version of this retrospective recorded CCH as wholly unimplemented. That was too strong, and the correction is instructive.

CCH's slogan was *reasoning as geodesic flow on a logical manifold*. The curl-free half died. **The geodesic half is live**, in a form the original did not anticipate: the substrate computes a Fisher-information metric over graph states, and `fisher_distance` measures **geodesic distance in that metric** between two cavities. For the Boltzmann form $p(G|\theta) = e^{-\beta E(G,\theta)}/Z(\theta)$, the Fisher metric is $g_{ij} = \beta^2 \,\mathrm{Cov}(\partial E/\partial\theta_i, \partial E/\partial\theta_j)$, estimated from per-tunnel co-firing correlations.

The manifold changed identity — from a continuous space of cognitive states to a **statistical manifold over graph configurations** — and the geodesic notion transferred intact. Two cavities at small Fisher distance are statistically indistinguishable, which makes them optimal analogical bridge candidates.

Status: curl-free constraint never implemented; geodesic framing LIVE on a different manifold. A paper can be half-dead, and recording it as wholly dead loses a real result.

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6. What Did Not Survive

Mechanism	Claim	Why it is dead
CMC	$O(1)$ training memory via a Cognitive State Buffer; $O(L \times H) \to O(H)$	Solves activation storage during **backward passes**. The current engine has no gradient training, so there is nothing to compress. Correct solution, vanished problem.
CMC — reversible dynamics	reconstruct activations by reversing the Hamiltonian flow instead of storing them	Depends on the dynamics being time-reversible. The current tick is explicitly **dissipative** — damping, Curie habituation, viscosity — so the trajectory cannot be run backwards even in principle.
CMC — zero optimizer state	no momentum or variance buffers, because ZeroGrad needs none	Not dead; **subsumed**. It is a corollary of §4.1 rather than a separate contribution.
BED	byte-level output by discrete argmin descent	Converged to the byte-mean; produced no prompt-dependent structure. A measured failure, not a deprecation.
CME	unified multimodal state without embeddings	Never implemented. Aspirational, and should never have been cited as a component.

Mechanism	Claim	Why it is dead
SCER	multi-mode routing at the reasoning layer	Subsumed by field-observable dispatch: the readout mode is chosen by arg-max over physical quantities, not by a router.
P-CMA transitions	learned transition operator	Subsumed by graph-rewriter transitions.

CMC is the most interesting entry: **the problem disappeared rather than the solution failing.** That is a distinct retirement reason worth recording separately — if gradient training ever returns to this lineage, CMC becomes relevant again unchanged, whereas BED and CME would not.

The reversible-dynamics row is the one genuine incompatibility. It is not that the mechanism is unneeded; it is that the successor substrate **forecloses** it, because dissipation is doing real work (thermal habituation is how ubiquitous concepts decouple themselves) and reversibility would have to be traded for it.

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7. Summary

Paper	Verdict	What survives
`zerograd.md`	**paper DEAD, rule LIVE**	local forward-only Hebbian update — foundational
`anq.md`	**scheme DEAD, pattern LIVE**	continuous integrator replacing threshold-commit
`CMA.md` (Conceptual Mutation)	**LIVE**	forced distant-pair hypothesis with decay on failure
`lto.md`	**binary DEAD, principle LIVE**	runtime acquisition with persistence
`rifa.md`	**LIVE**	Integrity Core, Silence Integration; misfiled
`cch.md`	**half dead**	curl-free never built; geodesic framing live on a statistical manifold
`hcma.md`	**retired, diagnosis survives**	collapse-from-global-signal; static potentials cannot sequence
`cma.md` (Cognitive Manifold)	**historical**	manifold/Hamiltonian vocabulary
`cmc.md`, `cmc_extra.md`	**problem vanished / foreclosed**	intact solution, no gradients left; reversibility incompatible with dissipation
`cme.md`	**never implemented**	nothing
`bed.md`	**measured failure**	negative result only
`scer.md`	**subsumed**	replaced by field-observable dispatch

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8. Recommendations for Citation

1. **Cite ZeroGrad, ANQ, and LTO as sources of mechanisms, never as descriptions of a system.** All three describe architectures that no longer exist and rules that are still running.
2. **Disambiguate CMA.** *Conceptual Mutation* is live; *Cognitive Manifold* is retired. Citing "CMA" unqualified is ambiguous in this lineage.
3. **Do not cite CCH's curl-free theorem as a property of any implemented engine.** It is a true theorem about a field that was never built. Its geodesic framing may be cited, on the statistical manifold.
4. **Cite HCMA and BED for their negative results.** They are the empirical basis for retiring an entire architectural family; deleting them would remove the evidence for that decision.
5. **Do not cite CME.** Nothing was built.
6. **Re-derive any parts list from source before repeating it.**

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9. Conclusion

The retired substrate failed as an architecture and succeeded as an experiment. It produced one large negative result — contrastive training on coherence yields a classifier, at every scale tested — and four mechanisms that outlived it.

The methodological point generalises past this lineage. Retirement should apply to proposals rather than ideas, and the only reliable way to tell which is which is to ask the source code rather than the literature. When the two disagree, as they did here in six places, the literature is wrong by construction: it was written earlier.

Two of those six were found only on a second pass — a live mechanism hidden behind a duplicated acronym, and a paper that was half alive. Neither would have been caught by reading abstracts.

References

- Mahir, I. *The Cognitive Manifold Architecture (CMA).* — retired.
- Mahir, I. *Conceptual Mutation Architecture (CMA).* — **live**; see `src/bisociation.hpp`.
- Mahir, I. *The Cognitive Continuum Hypothesis (CCH).*
- Mahir, I. *Hierarchical Cognitive Manifold Architecture (HCMA/DCHL).*
- Mahir, I. *The Cognitive Manifold Compressor (CMC).*
- Mahir, I. *ZeroGrad Cognitive Engine.* — dead paper, live rule.
- Mahir, I. *Adaptive Neural Quantisation (ANQ).* — dead scheme, live pattern.
- Mahir, I. *LTO: Live Training Over Manifolds.* — dead binary, live principle.
- Mahir, I. *Byte Energy Descent (BED).* — measured failure.
- Mahir, I. *Recursive Integrity Field Architecture (RIFA).* — live.
- Mahir, I. *A Boltzmann Fisher-Information Metric for Graph-Valued Cognitive States.* — where CCH's geodesic survived.
- Mahir, I. *From Artificial to Synthetic Intelligence.* §2, §5.5, §6.2.
- Hinton, G. *The Forward-Forward Algorithm*, 2022.